Internship Overview

Vegetation and Obstacle Perception

[ID 012026]

Develop a perception pipeline to estimate grass height detect invasive weeds and identify field obstacles for a small ground robot.

Objectives

- Estimate grass height in field conditions
- Detect invasive weeds in camera imagery
- Identify poles and other obstacles around the robot

Tasks

- Build image and ranging acquisition modules
- Process camera and laser data to estimate grass height
- Detect invasive weeds and mark detections
- Identify poles and similar obstacles
- Validate performance on the robot chassis

Requirements

- Python programming
- Computer vision fundamentals
- ROS basics

Equipment

- Jetson Orin Nano
- Teltonika RUT240 4G LTE Router
- VL53L0X ranging sensors
- Arducam IMX219 NoIR camera
- UGV Rover ROS 2 Open-source 6 Wheels 4WD

Deliverables

- Perception prototype for grass height weeds and obstacle detection
- Test results from field trials

Work Breakdown

- 20% hardware integration
- 80% coding

Duration: 2 months

Payment: Available in Bizquick

Supervisor: Kaleb Granados